

# CEO experience and firm risk-taking over business life cycle

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## Abstract

**Purpose** – This study aims to investigate how chief executive officer (CEO) characteristics, particularly managerial experience, affect the risk-taking behavior of listed firms in Vietnam.

**Design/methodology/approach** – Using a sample of 514 publicly listed firms from 2010 to 2021, the authors examine how corporate risk-taking measured by stock return volatility and idiosyncratic risk responds to CEO age, tenure and board experience.

**Findings** – The results indicate that CEO experience is inversely associated with corporate risk-taking. This effect is found consistently over different stages of business life cycle. Interestingly, highly educated CEOs are associated with lower levels of firm risk-taking during the decline stage.

**Research limitations/implications** – The findings support stewardship theory and underscore the importance of appointing highly experienced CEOs such as those with life, board and positional experience.

**Practical implications** – This study suggests that more experienced CEOs are associated with lower risk-taking, highlighting the importance of experience in CEO selection and succession planning. Firms operating in highly-uncertain environments or requiring greater risk control may benefit from prioritizing experienced CEOs to enhance stability.

**Originality/value** – This study extends the literature of corporate governance by contributing the evidence of the potential negative impact of CEO experience and CEO education on firm risk-taking.

**Keywords** Risk-taking, CEO experience, Life cycle, Stewardship theory, Vietnam

**Paper type** Research paper

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## 1. Introduction

Many aspects of organizational decision-making are influenced by manager traits, as documented by [Bertrand and Schoar \(2003\)](#), [Graham et al. \(2013\)](#), [Li and Tang \(2010\)](#) etc. Among managerial characteristics, chief executive officer (CEO) experience has been identified as one of the most important determinants of corporate outcomes. For instance, job-specific knowledge as well as extensive professional networks contribute to the positive effects of CEO work experience and age on firm performance ([Peni, 2014](#)). Moreover, CEOs with greater experience tend to adopt more proactive financial policies and allocate resources in ways that are less dependent on internal cash flows ([Custódio and Metzger, 2014](#)). [Hu and Liu \(2015\)](#) show that firms led by more experienced CEOs exhibit lower investment–cash flow sensitivity and make greater use of external financing sources, such as trade credit and bank loans. Furthermore, CEOs with higher education levels are more likely to be confident and proactive in decision-making, which can enhance their firms' creative performance ([Garcés-Galdeano and García-Olaverri, 2019](#)).

The impact of CEO experience on firms' risk-taking has been widely discussed in the literature. However, evidence from emerging markets remains limited, underscoring the need for further research in this context. Moreover, studies examining CEO experience often focus on exposure to extreme external events, such as famines and natural disasters (e.g.

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[Chen et al., 2021](#); [Long et al., 2020](#); [Bernile et al., 2017](#)). Such experiences, largely beyond a CEO's control, may enhance managers' crisis management skills, resilience and adaptability. In contrast, work experience refers to professional experience accumulated through employment, typically within specific industries, organizations or roles. This type of experience contributes to the development of managerial skills, career advancement and industry-specific knowledge of the managers.

The present study examines the influence of CEO work experience on firm risk-taking using a sample of 514 listed firms in Vietnam. Prior research (e.g. [Farak and Mallin, 2018](#); [Chen and Zheng, 2014](#); [Bsoul et al., 2022](#); [Serfling, 2014](#)) documents a significant relationship between CEO experience and corporate risk-taking; however, the direction of this effect remains inconclusive. Moreover, existing studies are largely concentrated on developed economies, such as the USA ([Chen and Zheng, 2014](#); [Serfling, 2014](#)) and the UK ([Wang and Poutziouris, 2010](#)), or primarily on major developing markets like China ([Farak and Mallin, 2018](#)). Vietnam offers a distinctive and underexplored setting for extending this line of research. According to the [World Bank \(2023\)](#), Vietnam is among the most dynamic emerging economies in East Asia, yet it exhibits institutional features that differ markedly from those of both developed economies and other emerging markets. These features include the legacy of a centrally planned economic system, high levels of state ownership, weak corporate governance, substantial information asymmetry and incomplete privatization ([OECD Vietnam, 2023](#); [KPMG Vietnam, 2024](#)). Despite ongoing reforms, state-owned enterprises (SOEs) continue to play a dominant role in the economy, suggesting that political influence may still shape firm-level decision-making.

These institutional features have important theoretical implications for understanding CEO behavior. Upper echelons theory posits that organizational outcomes reflect the cognitive bases and values of top executives ([Hambrick and Mason, 1984](#)). However, in the Vietnamese context, CEO discretion may be substantially constrained by political oversight associated with concentrated state ownership. As a result, the relationship between CEO experience and risk-taking may differ not only in magnitude but also in its underlying mechanisms. In particular, the influence of CEO characteristics on firm outcomes may be attenuated by restricted managerial discretion, especially in SOEs. At the same time, agency theory ([Jensen and Meckling, 1976](#)) emphasizes potential conflicts between managers and shareholders, particularly when the monitoring is weak. In Vietnam's relatively weak institutional environment, long-tenured or highly experienced CEOs may become entrenched, exploiting weak governance mechanisms to pursue either excessively risk-averse or self-serving strategies. This perspective suggests that CEO experience may either mitigate or exacerbate firm risk-taking, depending on ownership structure and governance conditions. Institutional theory further argues that the stronger institutional environment shapes the effectiveness of corporate governance mechanisms ([Meyer and Rowan, 1977](#); [Scott, 1987, 1995](#); [Oliver, 1996](#)). That means in such settings like Vietnam, compliance with regulatory or political norms may override personal preferences or strategic logic, especially in firms closely connected with the government. Consequently, CEO experience may play a more limited role in shaping firm risk-taking behavior.

In addition to contextualizing the CEO experience and risk-taking relationship, our study contributes to the literature by examining how this relationship varies across different business life cycle stages: introduction, growth, maturity and decline. While prior studies (e.g. [Habib and Hasan, 2017](#); [Shahzad et al., 2019](#)) show that firm risk-taking fluctuates across life cycle stages, few have explored how the impact of CEO traits on risk changes over time. We argue that experienced CEOs may play differentiated roles across life cycle phases, particularly during periods of organizational decline when strategic challenges intensify. Our results reveal that firms tend to take more risk during the decline stage, but CEOs with higher education and broader experience can help mitigate risk, potentially guiding firms back toward stability.

Overall, this study offers three contributions. First, it extends the literature of stewardship theory by providing evidence of the potential negative impact of CEO experience on firm risk-taking. Based on a dataset of firms listed in the Vietnamese market from 2010 to 2021, the study finds that CEO board experience, CEO tenure and CEO age are inversely related to the level at which a firm takes risk. This implies that experienced CEOs are reliable and accountable with company resources. Second, the impact is found to be stronger in firms managed by female CEOs and in non-SOEs. Notably, during the COVID-19 pandemic, the role of CEO experience is more pronounced. Finally, to the best of our knowledge, this is among the first studies investigating the relationship between CEO traits and risk-taking over the corporate life cycle, offering new insights into the dynamics of executive influence.

The rest of this paper is structured as follows. The related literature review is provided in Section 2. In Section 3, data collection and regression models are explained. In Section 4, research findings are presented. Our paper is concluded in Section 5.

## 2. Literature review

There are some theories that can be used to predict the influence of CEO experience on corporate risk-taking. The upper echelons theory ([Hambrick and Mason, 1984](#)) suggests that the traits of senior executives have a significant impact on the strategic decisions they make. This theory implies that companies led by older executives typically take less risk. Further, the theory predicts that highly educated CEOs are more receptive to new developments, whereas CEOs with prior professional experience are likely to be less receptive. Besides, the agency hypothesis ([Jensen and Meckling, 1976](#)) can be used to examine how CEOs affect firm risk-taking. According to this notion, agency conflicts occur when managers act in a way that does not serve the interests of shareholders. The agency theory implies that senior managers are opportunistic and self-benefit-seekers; thus, CEOs with more firm- and job-specific knowledge may make decisions that benefit their own interest (e.g. higher compensations or network expansion) rather than shareholder wealth. Conversely, according to the stewardship theory of [Davis et al. \(1997\)](#), managers are accountable, truthful and collectivist in their use of company resources. Consistent with this view, highly experienced and well-educated CEOs are more likely to focus on enhancing firm value and promoting sustainable growth, which is associated with lower levels of risk-taking.

In terms of empirical evidence, previous studies provide mixed evidence on the relationship of CEO experience and corporate risk-taking. On the one hand, [Farag and Mallin \(2018\)](#) find that young, short-tenured CEOs and CEOs with high board experience tend to take risky decisions. Accordingly, corporate risk-taking in Chinese firms is positively correlated with the percentage of directors on boards with postgraduate degrees. [Chen and Zheng \(2014\)](#), who also study CEO experience, discover a positive link between risk-taking and CEO tenure. They acknowledge that they cannot completely rule out explanations based on the power and experience effects of the tenure, but they suppose that tenure reflects managers' career concern and that the effect of CEO tenure on risk-taking depends on the asymmetry of information regarding the CEOs' ability. [Wang and Poutziouris \(2010\)](#) discover that CEO propensity for taking risk is positively related to their tenure.

In contrast, many studies find that longer tenure, older age and greater job experience are associated with increased risk aversion. [Bsoul et al. \(2022\)](#) observe a sample of 82 Jordanian listed firms and find a significant negative relationship between CEO experience and risk-taking. [Serfling \(2014\)](#) also documents a negative association between stock return volatility and CEO age. He explains that older CEOs take less risk since they face lower career concerns compared with younger managers. Similarly, [Bertrand and Schoar \(2003\)](#) demonstrate that old CEOs tend to be more conservative when making decisions. Additionally, a longer stay in management positions denotes more entrenchment and management influence. To shield their own private benefits, long-tenured managers could be less willing to take risk ([Bertrand and Mullainathan, 2003](#)). Consistently, [Orens and Reheul \(2013\)](#) claim that

long-tenured, highly experienced and older CEOs are higher risk-aversion than short-tenured and newly appointed CEOs. They also suppose that highly educated managers typically focus on long-term sustainable development rather than high-variance projects, which might result in significant losses. Wang and Poutziouris (2010) discover that CEO propensity for taking risk is negatively related to CEO age but not connected to the education background of CEOs. Since there are mixed evidence on the influence of CEO experience and firm risk-taking, we propose our first hypothesis as follows:

*H1a.* CEO experience is negatively related to the level of risk-taking of Vietnamese listed firms.

*H1b.* CEO experience is positively related to the level of risk-taking of Vietnamese listed firms.

According to the corporate life cycle theory (see Miller and Friesen, 1984; Quinn and Cameron, 1983), at each stage of life, firms use different business strategies. They also have many changes in asset and capital structures; as a result, shifts in their risk-taking profiles can be anticipated. Habib and Hasan (2017) show that risk-taking level is high in the stages of introduction and decline, but low in growth and maturity phases. They explain that during the first phase of the life, firms pursue risky investment to gain market shares and expand the operation, and in the last stage, the aim to back to profitability again also drives firms to take more risk. Shahzad *et al.* (2019), consistently, find that risk-taking level is higher in the stages of introduction and decline, while lower in the two middle phases. Based on these previous evidences, we propose the second hypothesis as follows:

*H2.* The relationship between CEO experience and risk-taking changes across life cycle stages.

### 3. Methodology

#### 3.1 Data

Our data set comprises 4,349 firm-year observations from 514 firms listed on the Ho Chi Minh and Hanoi stock exchanges over the period from 2010 to 2021. The data are unique and comprehensive, with CEO information collected manually and firm-level financial data obtained from the FiinPro database. In line with prior studies in this topic (e.g. Chen and Zheng, 2014; Armstrong and Vashishtha, 2012; Farag and Mallin, 2018, etc.), firms in the financial and utilities sectors are excluded due to differences in accounting practice, operational structures, governance characteristics and objectives. To mitigate the influence of outliers, all variables are winsorized at the 1<sup>st</sup> and 99<sup>th</sup> percentiles.

#### 3.2 Model

The baseline model is as follows:

$$Risk\_taking_{it} = \alpha_0 + \alpha_1 CEO\_experience_{it-1} + \alpha_2 Education_{it-1} + \delta Controls_{it-1} + \varepsilon_{it} \quad (1)$$

where Risk\_taking is measured by stock return volatility (STD) and idiosyncratic risk (IDIO), respectively, as suggested by previous studies on the same topic including Farag and Mallin (2018), Chen and Zheng (2014). Stock return volatility is the annualized standard deviation of daily stock return. Idiosyncratic risk is the annualized standard deviation of residuals estimated from equation (2) in a given year:

$$Stock\ Return_{it} = \alpha + \beta_i Market\_Return_t + \varepsilon_{it} \quad (2)$$

Note that the market return is measured as the daily return of the VN-index.

CEO\_experience and Education are CEO experience and CEO education, respectively. CEO experience is measured by three alternative metrics, including the natural logarithm of CEO age which serves as an indicator of life experience (Age); the natural logarithm of the number of years the CEO has held their current position (Tenure); and the natural logarithm of the number of years of experience they have accumulated on the board (Board\_experience). These measures of CEO experience have been widely used in prior research, including Wang *et al.* (2016), Orens and Reheul (2013), Bertrand and Schoar (2003) and Farag and Mallin (2018), which are supposed to provide a comprehensive view of CEOs' life, career and board-related experiences. Each of these experience measures is analyzed separately for two main reasons. First, these measures are highly correlated with one another, thus analyzing them individually allows us to better isolate their distinct effects on firm risk-taking and mitigate concerns about multicollinearity. Second, our approach follows the methodology adopted in previous studies in the field, such as the works of Wang *et al.* (2016) and Orens and Reheul (2013), ensuring the consistency with established research practices.

In the equation (1), we control for a set of firm characteristics that can affect corporate risk-taking level, including firm size (Firm\_size), market-to-book ratio (MTB) as a proxy of investment opportunities, asset tangibility (Tang), growth in sales (Sales), leverage (Lev) and market return (RMKT). According to existing evidence of corporate risk-taking provided by Peltomäki *et al.* (2021), Coles *et al.* (2006), Armstrong and Vashishtha (2012) and Yung and Chen (2018), they are important risk-taking determinants. Definitions of all variables are given in Appendix 1. We also include industry and year fixed effects in all regressions.

To further ensure the robustness of results, the expansion version of equation (1), added three control variables denoted to board characteristics, including duality (Duality), board size (Board\_size) and independent ratio of the board (Independent), is estimated additionally.

Next, to test the influence of firm life cycle on the relationship between CEO traits and corporate risk-taking, we add into the equation (1) dummy variables denoted to stages of business life:

$$\begin{aligned} \text{Risk\_taking}_{it} = & \alpha_0 + \alpha_1 \text{CEO\_experience}_{it-1} + \alpha_2 \text{Education}_{it-1} + \beta \text{Stage\_dummy}_{it-1} \\ & + \gamma \text{CEO\_experience}_{it-1} * \text{Stage\_dummy}_{it-1} \\ & + \theta \text{Education}_{it-1} * \text{Stage\_dummy}_{it-1} + \delta \text{Controls}_{it-1} + \varepsilon_{it} \end{aligned} \quad (3)$$

where Stage\_dummy is a vector that captures the different stages of a firm's life cycle, including (1) Growth\_dummy, which takes the value of 1 if a firm is in the growth phase, and 0 otherwise; (2) Maturity\_dummy, which takes the value of 1 if a firm is in the maturity phase and 0 otherwise; (3) Decline\_dummy, which takes the value of 1 if a firm is in the decline phase, and 0 otherwise. Since there are four stages of the business life, three dummy variables are included.

## 4. Results

### 4.1 Summary statistics

Table 1 provides summary statistics for all main variables. As shown, the mean of STD and IDIO are about 0.45 and 0.43, respectively. CEOs in our sample, on average, are 49.61 years old, have 15.97 years on boards and 6.63 years in executive roles [1]. Most observed firms are managed by CEOs with a bachelor degree and higher.

The pairwise correlation coefficients are reported in Table 2. As can be seen, the three measures of CEO experience are negatively correlated with the two risk measures. Note that Age, Tenure and Board\_experience have high correlation coefficients (0.423, 0.480 and 0.574); as such, we estimate equation (1) with these three variables separately to avoid the problem of multicollinearity [2].

| Table 1 Summary  |       |         |        |         |         |
|------------------|-------|---------|--------|---------|---------|
| Variable         | Obs   | Mean    | SD     | Min.    | Max.    |
| STD              | 4,349 | 0.4476  | 0.1465 | 0.0000  | 1.0635  |
| IDIO             | 4,349 | 0.4338  | 0.1492 | 0.0000  | 1.0582  |
| Age              | 3,856 | 3.8904  | 0.1661 | 3.4340  | 4.2195  |
| Board_experience | 3,860 | 2.5872  | 0.8371 | 0.0000  | 3.6376  |
| Tenure           | 3,850 | 1.6690  | 0.9228 | 0.0000  | 3.2958  |
| Education        | 3,855 | 2.2446  | 0.4779 | 1.0000  | 3.0000  |
| Duality          | 3,861 | 0.2458  | 0.4306 | 0.0000  | 1.0000  |
| Board_size       | 3,861 | 5.5284  | 1.1506 | 3.0000  | 9.0000  |
| Independent      | 3,861 | 0.1618  | 0.2171 | 0.0000  | 1.0000  |
| Firm_size        | 4,349 | 27.2388 | 1.5529 | 23.6791 | 31.5847 |
| MTB              | 4,348 | 0.7893  | 0.6557 | -0.4002 | 8.7283  |
| Tang             | 4,349 | 0.2084  | 0.1983 | 0.0000  | 0.8562  |
| Sales            | 4,342 | 0.1108  | 0.4868 | -0.7132 | 3.6698  |
| Lev              | 4,349 | 0.2187  | 0.1866 | 0.0000  | 0.6909  |
| RMKT             | 4,349 | 0.1447  | 0.1323 | -0.0978 | 0.3923  |

**Note(s):** The table presents descriptive statistics of all variables. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

## 4.2 CEO experience and firm risk-taking

The regression results for [equation \(1\)](#) are presented in [Table 3](#). As shown, all experience measures are negatively related to the two measures of risk-taking. First, the results in columns 1 and 2 suggest that older CEOs typically have greater experience managing stock price volatility. In an Asian market such as Vietnam, older CEOs may exert greater influence over corporate boards and use their experience to improve company governance. This finding is consistent with the results of [Bertrand and Schoar \(2003\)](#), [Orens and Reheul \(2013\)](#) and [Wang and Poutziouris \(2010\)](#) which indicate that older CEOs are typically more risk-averse. Further, the time CEOs serve on boards is negatively associated with risk-taking, which is shown by our second proxy for CEO experience (see columns 3 and 4). This finding is reasonable because CEOs with prior board experience are more knowledgeable about internal operations, company culture and the external environment of their companies. Finally, longer CEO tenure is inversely associated with risk-taking (see columns 5 and 6). This result supports the claims made by those arguing that CEOs with longer tenure: (1) are more committed to their vision and goals for a company's management (e.g. [Hambrick and Fukutomi, 1991](#)); (2) have broader personal networks (e.g. [Finkelstein and Hambrick, 1996](#)); (3) give the organizations legitimacy (e.g. [Orens and Reheul, 2013](#)); and (4) have more influence (e.g. [Chen and Zheng, 2014](#)). In short, the results support our *H1a*.

[Equation \(1\)](#) is re-estimated with the inclusion of additional control variables related to board characteristics, including board size, board independence and CEO duality. The results reported in [Table 4](#) are consistent with the previous findings and continue to support *H1a*.

Previous studies show that, compared to men, women is less willing to take risk ([Schwartz, 1989](#); [Jianakoplos and Bernasek, 1998](#)). The main reason may be lie on the trait of greater risk aversion, which is often observed in women in a variety of decision-making contexts ([Jianakoplos and Bernasek, 1998](#); [Schwartz, 1989](#)). Furthermore, compared to men, women in leadership roles are more vulnerable to dangers and hazards ([Ryan and Haslam, 2007](#)). Female CEOs are more careful in their choices since they are more likely to be fired than their male counterparts ([Gupta et al., 2020](#)).

To determine whether there are differences between female and male directors, we investigate the association between CEO experience and risk-taking for two subsets. [Table 5](#) indicates that companies led by female CEOs have larger CEO experience coefficient magnitudes.

**Table 2** Correlation matrix

| No. Variables      | 1         | 2         | 3         | 4         | 5         | 6         | 7         | 8        | 9       | 10       | 11       | 12        | 13     | 14    | 15 |
|--------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------|---------|----------|----------|-----------|--------|-------|----|
| 1 STD              | 1         |           |           |           |           |           |           |          |         |          |          |           |        |       |    |
| 2 IDIO             | 0.987***  | 1         |           |           |           |           |           |          |         |          |          |           |        |       |    |
| 3 Age              | -0.110*** | -0.106*** | 1         |           |           |           |           |          |         |          |          |           |        |       |    |
| 4 Board_experience | -0.142*** | -0.145*** | 0.480**   | 1         |           |           |           |          |         |          |          |           |        |       |    |
| 5 Tenure           | -0.070*** | -0.071*** | 0.423***  | 0.574***  | 1         |           |           |          |         |          |          |           |        |       |    |
| 6 Education        | -0.025    | -0.034*   | -0.140*** | -0.067*** | -0.079*** | 1         |           |          |         |          |          |           |        |       |    |
| 7 Duality          | 0.028     | 0.028     | 0.185***  | 0.205***  | 0.350***  | -0.069*** | 1         |          |         |          |          |           |        |       |    |
| 8 Board_size       | -0.155*** | -0.182*** | 0.092**   | 0.077***  | 0.052**   | -0.023    | -0.020    | 1        |         |          |          |           |        |       |    |
| 9 Independent      | -0.042**  | -0.043**  | -0.094**  | -0.068*** | -0.039*   | -0.029    | -0.015    | -0.021   | 1       |          |          |           |        |       |    |
| 10 Firm_size       | -0.280*** | -0.341*** | -0.005    | 0.030     | -0.011    | 0.133***  | -0.080*** | 0.343*** | -0.039* | 1        |          |           |        |       |    |
| 11 MTB             | -0.228*** | -0.230*** | 0.106***  | 0.076***  | 0.044**   | -0.025    | -0.021    | 0.189*** | 0.028   | 0.154*** | 1        |           |        |       |    |
| 12 Tang            | -0.118*** | -0.119*** | 0.056**   | 0.051**   | -0.029    | -0.010    | -0.096*** | 0.120*** | 0.009   | 0.126*** | 0.191*** | 1         |        |       |    |
| 13 Sales           | 0.027     | 0.029     | -0.062*** | -0.050**  | -0.005    | -0.025    | 0.050**   | 0.010    | -0.009  | 0.041*   | 0.000    | -0.055*** | 1      |       |    |
| 14 Lev             | -0.089*** | -0.106*** | -0.030    | -0.019    | 0.001     | 0.036*    | -0.011    | 0.101*** | -0.008  | 0.422*** | -0.026   | 0.246***  | 0.009  | 1     |    |
| 15 RMKT            | -0.037*   | -0.029    | -0.011    | 0.007     | -0.007    | -0.001    | 0.006     | -0.007   | -0.004  | -0.028   | 0.015    | 0.002     | -0.002 | 0.006 | 1  |

**Note(s):** The table presents pairwise correlation coefficients. \*, \*\* and \*\*\* indicate statistical significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

**Table 3** The relationship between CEO experience, education and firm risk-taking

| Variables        | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                |
|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Age              | −0.0811*** (0.0212) | −0.0788*** (0.0216) | −0.0221*** (0.0045) | −0.0222*** (0.0045) | −0.0113*** (0.0034) | −0.0117*** (0.0034) |
| Board_experience |                     |                     |                     |                     |                     |                     |
| Tenure           |                     |                     |                     |                     |                     |                     |
| Education        | −0.0035 (0.0074)    | −0.0039 (0.0075)    | −0.0024 (0.0073)    | −0.0029 (0.0075)    | −0.0015 (0.0074)    | −0.0021 (0.0075)    |
| Firm_size        | −0.0270*** (0.0032) | −0.0343*** (0.0032) | −0.0262*** (0.0031) | −0.0335*** (0.0032) | −0.0271*** (0.0032) | −0.0344*** (0.0033) |
| MTB              | −0.0388*** (0.0057) | −0.0394*** (0.0058) | −0.0383*** (0.0058) | −0.0390*** (0.0059) | −0.0406*** (0.0059) | −0.0411*** (0.0061) |
| Tang             | −0.0315 (0.0211)    | −0.0289 (0.0216)    | −0.0305 (0.0212)    | −0.0278 (0.0217)    | −0.0362* (0.0214)   | −0.0336 (0.0219)    |
| Sales            | 0.0109* (0.0051)    | 0.0098* (0.0050)    | 0.0103* (0.0050)    | 0.0092* (0.0049)    | 0.0120** (0.0051)   | 0.0109** (0.0051)   |
| Lev              | 0.0342 (0.0233)     | 0.0414* (0.0234)    | 0.0315 (0.0231)     | 0.0387* (0.0232)    | 0.0385* (0.0232)    | 0.0458* (0.0233)    |
| RMKT             | 0.1764*** (0.0154)  | 0.2169*** (0.0153)  | 0.1658*** (0.0152)  | 0.2061*** (0.0152)  | 0.1760*** (0.0154)  | 0.2160*** (0.0153)  |
| Constant         | 1.5666*** (0.1216)  | 1.7320*** (0.1234)  | 1.2820*** (0.0847)  | 1.4569*** (0.0862)  | 1.2683*** (0.0867)  | 1.4443*** (0.0883)  |
| Observations     | 4,349               | 4,349               | 4,352               | 4,352               | 4,340               | 4,340               |
| R-squared        | 0.1759              | 0.2039              | 0.1826              | 0.2108              | 0.1751              | 0.2038              |

**Note(s):** The table shows the regression results of [equation \(1\)](#). CEO experience is measured by CEO age (Age), time on board of the CEO (Board\_experience) and CEO tenure (Tenure) while risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Control variables include firm size (Firm\_size), Market-to-book ratio (MTB), Tangibility (Tang), Sales growth (Sales), Leverage (Lev) and Market return (RMKT). Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\* and \*\*\* indicate significance at the 10%, 5% and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

**Table 4** Additional control variables

| Variables        | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                |
|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Age              |                     |                     |                     |                     |                     |                     |
| Board_experience | −0.0896*** (0.0213) | −0.0864*** (0.0216) | −0.0239** (0.0045)  | −0.0238** (0.0046)  | −0.0135*** (0.0036) | −0.0138*** (0.0036) |
| Tenure           |                     |                     |                     |                     |                     |                     |
| Education        | −0.0043 (0.0074)    | −0.0048 (0.0075)    | −0.0029 (0.0073)    | −0.0035 (0.0074)    | −0.0020 (0.0074)    | −0.0027 (0.0075)    |
| Duality          | 0.0097 (0.0077)     | 0.0083 (0.0079)     | 0.0126* (0.0076)    | 0.0115 (0.0078)     | 0.0135 (0.0083)     | 0.0126 (0.0084)     |
| Board_size       | −0.0041 (0.0033)    | −0.0047 (0.0034)    | −0.0041 (0.0033)    | −0.0048 (0.0034)    | −0.0042 (0.0034)    | −0.0048 (0.0035)    |
| Independent      | −0.0384** (0.0171)  | −0.0389** (0.0175)  | −0.0372* (0.0165)   | −0.0379** (0.0169)  | −0.0327** (0.0166)  | −0.0334** (0.0170)  |
| Firm_size        | −0.0259*** (0.0034) | −0.0330*** (0.0035) | −0.0249*** (0.0034) | −0.0320*** (0.0035) | −0.0258*** (0.0035) | −0.0330*** (0.0036) |
| MTB              | −0.0378*** (0.0059) | −0.0384*** (0.0060) | −0.0374*** (0.0060) | −0.0379*** (0.0061) | −0.0396*** (0.0061) | −0.0401*** (0.0062) |
| Tang             | −0.0275 (0.0212)    | −0.0247 (0.0217)    | −0.0260 (0.0212)    | −0.0230 (0.0217)    | −0.0324 (0.0215)    | −0.0296 (0.0219)    |
| Sales            | 0.0104* (0.0050)    | 0.0094* (0.0050)    | 0.0098* (0.0050)    | 0.0088* (0.0049)    | 0.0116** (0.0051)   | 0.0105** (0.0050)   |
| Lev              | 0.0319 (0.0235)     | 0.0390* (0.0236)    | 0.0288 (0.0233)     | 0.0359 (0.0235)     | 0.0364 (0.0234)     | 0.0436* (0.0235)    |
| RMKT             | 0.1805*** (0.0163)  | 0.2197*** (0.0163)  | 0.1711*** (0.0161)  | 0.2105*** (0.0161)  | 0.1822*** (0.0163)  | 0.2213*** (0.0163)  |
| Constant         | 1.5961*** (0.1244)  | 1.7582*** (0.1262)  | 1.2783*** (0.0865)  | 1.4529*** (0.0881)  | 1.2640*** (0.0885)  | 1.4395*** (0.0903)  |
| Observations     | 4,349               | 4,349               | 4,352               | 4,352               | 4,340               | 4,340               |
| R-squared        | 0.1806              | 0.2086              | 0.1877              | 0.2160              | 0.1797              | 0.2084              |

**Note(s):** The table shows the regression results of [equation \(1\)](#). CEO experience is measured by CEO age (Age), time on board of the CEO (Board\_experience) and CEO tenure (Tenure) while risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Control variables include board size (Board\_size), Duality (Duality), Independent (Independent), firm size (Firm\_size), Market-to-book ratio (MTB), Tangibility (Tang), Sales growth (Sales), Leverage (Lev) and Market return (RMKT). Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\*, and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

**Table 5** CEO gender

| Variables        | Female CEO         |                     |                     |                     |                    |                     |
|------------------|--------------------|---------------------|---------------------|---------------------|--------------------|---------------------|
|                  | STD                | IDIO                | STD                 | IDIO                | STD                | IDIO                |
| Age              | −0.1970** (0.0821) | −0.2084** (0.0881)  | −0.0432*** (0.0139) | −0.0460** (0.0142)  | −0.0254* (0.0131)  | −0.0283** (0.0136)  |
| Board_experience |                    |                     |                     |                     | 0.0028 (0.0330)    | 0.0047 (0.0342)     |
| Tenure           |                    |                     |                     |                     | −0.0221** (0.0092) | −0.0294*** (0.0096) |
| Education        | −0.0039 (0.0262)   | −0.0019 (0.0276)    | 0.0194 (0.0303)     | 0.0228 (0.0314)     | −0.0224 (0.0144)   | −0.0236 (0.0149)    |
| Firm_size        | −0.0261** (0.0084) | −0.0339*** (0.0087) | −0.0201** (0.0096)  | −0.0274*** (0.0100) | −0.0209 (0.0672)   | 0.0070 (0.0706)     |
| MTB              | −0.0151 (0.0113)   | −0.0160 (0.0118)    | −0.0185 (0.0132)    | −0.0195 (0.0135)    | −0.0098 (0.0189)   | −0.0121 (0.0196)    |
| Tang             | −0.0434 (0.0586)   | −0.0159 (0.0619)    | −0.0139 (0.0607)    | 0.0152 (0.0640)     | 0.0738 (0.0790)    | 0.0807 (0.0807)     |
| Sales            | −0.0088 (0.0172)   | −0.0108 (0.0183)    | −0.0166 (0.0180)    | −0.0192 (0.0184)    | 0.1602** (0.0642)  | 0.1784*** (0.0647)  |
| Lev              | 0.0512 (0.0848)    | 0.0579 (0.0880)     | 0.0676 (0.0806)     | 0.0750 (0.0834)     | 1.1025*** (0.2795) | 1.2725*** (0.2896)  |
| RMKT             | 0.1585** (0.0619)  | 0.1779*** (0.0624)  | 0.1182* (0.0675)    | 0.1348* (0.0678)    | 306                | 306                 |
| Constant         | 1.9297*** (0.3309) | 2.1510*** (0.3606)  | 1.0708*** (0.2836)  | 1.2417*** (0.2934)  | 0.2714             | 0.3086              |
| Observations     | 306                | 306                 | 306                 | 306                 |                    |                     |
| R-squared        | 0.2945             | 0.3302              | 0.3080              | 0.3453              |                    |                     |

**Note(s):** The table shows the regression results of [equation \(1\)](#) for two subsamples: firms managed by female vs by male CEOs. CEO experience is measured by CEO age (Age), time on board of the CEO (Board\_experience) and CEO tenure (Tenure) while risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\* and \*\*\* indicate significance at the 10%, 5% and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

(continued)

**Note(s):** The table shows the regression results of [equation \(1\)](#) for two subsamples: firms managed by female vs by male CEOs. CEO experience is measured by CEO age (Age), time on board of the CEO (Board\_experience) and CEO tenure (Tenure) while risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\* and \*\*\* indicate significance at the 10%, 5% and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

(continued)

**Table 5**

| Variables        | STD                 | IDIO                | STD                 | Male CEO            | IDIO                | STD                 | IDIO                |
|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Age              | −0.0713*** (0.0222) | −0.0672** (0.0222)  | −0.0207*** (0.0046) | −0.0206*** (0.0047) |                     |                     |                     |
| Board_experience |                     |                     |                     |                     |                     |                     |                     |
| Tenure           |                     |                     |                     |                     |                     | −0.0107*** (0.0035) | −0.0110*** (0.0036) |
| Education        | −0.0030 (0.0078)    | −0.0033 (0.0079)    | −0.0024 (0.0077)    | −0.0028 (0.0078)    | −0.0009 (0.0077)    |                     | −0.0015 (0.0078)    |
| Firm_size        | −0.0275*** (0.0034) | −0.0349*** (0.0034) | −0.0269*** (0.0033) | −0.0342*** (0.0033) | −0.0278*** (0.0034) |                     | −0.0351*** (0.0034) |
| MTB              | −0.0422*** (0.0062) | −0.0428*** (0.0063) | −0.0420*** (0.0062) | −0.0425*** (0.0064) | −0.0442*** (0.0063) |                     | −0.0447*** (0.0064) |
| Tang             | −0.0301 (0.0224)    | −0.0293 (0.0229)    | −0.0291 (0.0225)    | −0.0280 (0.0230)    | −0.0346 (0.0226)    |                     | −0.0337 (0.0231)    |
| Sales            | 0.0133** (0.0054)   | 0.0124* (0.0054)    | 0.0130** (0.0053)   | 0.0120** (0.0052)   | 0.0145*** (0.0055)  |                     | 0.0135** (0.0054)   |
| Lev              | 0.0346 (0.0242)     | 0.0419* (0.0243)    | 0.0314 (0.0241)     | 0.0388 (0.0242)     | 0.0379 (0.0242)     |                     | 0.0454* (0.0244)    |
| RMKT             | 0.1817*** (0.0163)  | 0.2235*** (0.0162)  | 0.1725*** (0.0161)  | 0.2143*** (0.0160)  | 0.1812*** (0.0163)  |                     | 0.2226*** (0.0162)  |
| Constant         | 1.5454*** (0.1305)  | 1.7042*** (0.1309)  | 1.3019*** (0.0904)  | 1.4768*** (0.0911)  | 1.2890*** (0.0919)  |                     | 1.4653*** (0.0926)  |
| Observations     | 4,043               | 4,043               | 4,046               | 4,046               | 4,034               |                     | 4,034               |
| R-squared        | 0.1723              | 0.1998              | 0.1788              | 0.2067              | 0.1734              |                     | 0.2016              |

Hence, the presence of female CEOs in companies may result in greater risk-taking-reducing impacts of CEO experience. To ensure the significance of difference in CEO experience coefficients, we run Chow test and present the results on [Appendix 2](#). As shown in the panel A of the [Appendix 2](#), the differences in the impact of CEO experience on firm risk-taking between firms managed by male and female CEOs are statistically significant.

Next, given the fact that SOEs (i.e. the government controls more than 50% of the firm shares) account for about 29% of our sample [\[3\]](#), we investigate the relationship between CEO experience and risk-taking for two subsets: non-SOEs [\[4\]](#) and SOEs, to explore the difference between the two types of firms. According to [Jin et al. \(2024\)](#), state ownership greatly increases businesses' ability to take risk by facilitating resource access and lowering the regulatory requirements. However, [Boubakri et al. \(2013\)](#) provide evidence that corporate risk-taking is adversely correlated with state ownership. The authors explain that state-controlled firms are established to serve political interests, not to seek higher performance so they are less likely to undertake risky decisions. These evidences suggest us to separate non-SOEs from SOEs when studying risk-taking topic.

[Table 6](#) presents the influence of CEO experience on risk-taking for the two groups of SOEs and non-SOEs. Besides, Panel B of [Appendix 2](#) shows the Chow test's results to explore the significance of difference in CEO experience coefficients between the two groups. As demonstrated, SOEs and non-SOEs differ significantly in how CEO experience affects business risk-taking. The presence of highly experienced CEOs can reduce risk-taking in non-SOEs but not in SOEs. It may show that managers in state-controlled companies may not have sufficient decision-making power.

The COVID-19 pandemic has leaved undeniable effects on several corporate aspects, including risk (see [Nasih et al., 2024](#); [Yang et al., 2025](#)). Given that the research period runs from 2020 to 2021, it is necessary to check whether the pandemic altered the nature of the association between CEO experience and risk-taking. [Table 7](#) shows regression results for the non-crisis (before 2020) and COVID-19 (2020 and 2021) periods. As can be seen, the detrimental effects of a CEO's experience occur across two separate time periods. It is interesting to note that, during the pandemic, CEO experience seems to matter more. To know whether the difference in CEO experience coefficients are statistically significant, we run Chow test and present the results on Panel C of [Appendix 2](#). As shown, there is a significant difference in the influence of CEO age, CEO board experience and tenure on firm risk-taking between pre- and during the COVID-19 period.

We must be aware of some potential sources of endogeneity. For example, companies may hire CEOs whose risk preferences align with their appetite for risk-taking. For example, it is possible that firms with inherently higher risk appetites or strategic orientations toward innovation and growth appoint younger CEOs who are more willing to take risk, as suggested by [Bertrand and Schoar \(2003\)](#). This could lead to reverse-causality, where firm risk-taking influences the type of CEO chosen, rather than CEO experience affecting firm risk-taking. Besides, endogeneity can arise from omitted variables (e.g. corporate culture may encourage or discourage risk-taking, regardless of the CEO experience) and measurement errors (e.g. years working at the company may not reflect the whole experience of CEOs). These issues can lead to bias in investigating the relationship between CEO experience and risk-taking.

To mitigate endogeneity, we re-estimate the [equation \(1\)](#) by the method of two-step system generalized method of moments (i.e. system-GMM) proposed by [Arellano and Bond \(1991\)](#) and [Blundell and Bond \(1998\)](#). This dynamic panel data model uses one-year lags of explanatory variables, including CEO experience and other control variables, as instruments. The [Hansen \(1982\)](#) test statistics for overidentifying restriction is used to test the validity of the instruments. Besides, AR2 test for second-order autocorrelations is conducted.

**Table 6** SOEs and non-SOEs

| Variables        | STD                 | IDIO                | STD                 | SOE | IDIO                | STD                 | IDIO                |
|------------------|---------------------|---------------------|---------------------|-----|---------------------|---------------------|---------------------|
| Age              | -0.0362 (0.0462)    | -0.0380 (0.0464)    | -0.0126* (0.0074)   |     | -0.0124 (0.0078)    |                     |                     |
| Board_experience |                     |                     |                     |     |                     | -0.0164*** (0.0062) | -0.0162** (0.0063)  |
| Tenure           |                     |                     |                     |     |                     | 0.0076 (0.0132)     | 0.0062 (0.0133)     |
| Education        | 0.0060 (0.0138)     | 0.0046 (0.0139)     | 0.0056 (0.0132)     |     | 0.0043 (0.0133)     |                     |                     |
| Firm_size        | -0.0283*** (0.0054) | -0.0367*** (0.0053) | -0.0279*** (0.0055) |     | -0.0363*** (0.0054) | -0.0288*** (0.0054) | -0.0372** (0.0053)  |
| MTB              | -0.0378*** (0.0076) | -0.0386*** (0.0075) | -0.0379*** (0.0079) |     | -0.0386*** (0.0078) | -0.0379*** (0.0079) | -0.0386*** (0.0078) |
| Tang             | 0.0239 (0.0321)     | 0.0254 (0.0334)     | 0.0249 (0.0323)     |     | 0.0264 (0.0336)     | 0.0207 (0.0320)     | 0.0223 (0.0333)     |
| Sales            | 0.0068 (0.0133)     | 0.0059 (0.0146)     | 0.0067 (0.0132)     |     | 0.0058 (0.0145)     | 0.0074 (0.0134)     | 0.0066 (0.0147)     |
| Lev              | 0.0431 (0.0394)     | 0.0570 (0.0398)     | 0.0397 (0.0392)     |     | 0.0537 (0.0397)     | 0.0435 (0.0388)     | 0.0573 (0.0392)     |
| RMKT             | 0.1833*** (0.0309)  | 0.2275*** (0.0307)  | 0.1760*** (0.0310)  |     | 0.2202*** (0.0309)  | 0.1794*** (0.0308)  | 0.2236*** (0.0305)  |
| Constant         | 1.3830*** (0.2356)  | 1.6000*** (0.2302)  | 1.2619*** (0.1429)  |     | 1.4720*** (0.1389)  | 1.2731*** (0.1445)  | 1.4824*** (0.1404)  |
| Observations     | 1,246               | 1,246               | 1,246               |     | 1,246               | 1,245               | 1,245               |
| R-squared        | 0.1679              | 0.2097              | 0.1710              |     | 0.2125              | 0.1757              | 0.2167              |

**Note(s):** The table shows the regression results of [equation \(1\)](#) for two subsamples: state-owned and non-state-owned enterprises. CEO experience is measured by CEO age (Age), time on board of the CEO (Board\_experience) and CEO tenure (Tenure) while risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\* and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

**Note(s):** The table shows the regression results of [equation \(1\)](#) for two subsamples: state-owned and non-state-owned enterprises. CEO experience is measured by CEO age (Age), time on board of the CEO (Board\_experience) and CEO tenure (Tenure) while risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\*, and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

(continued)

**Table 6**

| Variables        | Non-SOE             |                     |                     |                     |                     |                     |
|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                  | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                |
| Age              | -0.0913*** (0.0232) | -0.0887*** (0.0237) | -0.0248*** (0.0054) | -0.0249*** (0.0054) | -0.0106*** (0.0040) | -0.0110*** (0.0040) |
| Board_experience |                     |                     |                     |                     | -0.0072 (0.0090)    | -0.0078 (0.0092)    |
| Tenure           | -0.0100 (0.0088)    | -0.0104 (0.0090)    | -0.0075 (0.0089)    | -0.0081 (0.0091)    | -0.0268*** (0.0038) | -0.0340*** (0.0039) |
| Education        | -0.0273*** (0.0038) | -0.0344*** (0.0039) | -0.0262*** (0.0037) | -0.0333*** (0.0038) | -0.0411*** (0.0078) | -0.0415*** (0.0080) |
| Firm_size        | -0.0381*** (0.0073) | -0.0385*** (0.0074) | -0.0374*** (0.0073) | -0.0378*** (0.0074) |                     |                     |
| MTB              |                     |                     |                     |                     |                     |                     |
| Tang             | -0.0634** (0.0272)  | -0.0619** (0.0275)  | -0.0604** (0.0274)  | -0.0587** (0.0276)  | -0.0660** (0.0280)  | -0.0644** (0.0283)  |
| Sales            | 0.0095* (0.0053)    | 0.0088* (0.0053)    | 0.0086* (0.0052)    | 0.0078 (0.0051)     | 0.0106* (0.0054)    | 0.0097* (0.0053)    |
| Lev              | 0.0255 (0.0281)     | 0.0316 (0.0282)     | 0.0220 (0.0279)     | 0.0281 (0.0280)     | 0.0294 (0.0283)     | 0.0356 (0.0284)     |
| RMKT             | 0.1739*** (0.0184)  | 0.2130*** (0.0183)  | 0.1629*** (0.0181)  | 0.2018*** (0.0180)  | 0.1739*** (0.0185)  | 0.2123*** (0.0184)  |
| Constant         | 1.6397*** (0.1381)  | 1.7959*** (0.1421)  | 1.3157*** (0.1000)  | 1.4828*** (0.1029)  | 1.2900*** (0.1030)  | 1.4583*** (0.1061)  |
| Observations     | 3,103               | 3,103               | 3,106               | 3,106               | 3,095               | 3,095               |
| R-squared        | 0.1895              | 0.2118              | 0.1974              | 0.2203              | 0.1865              | 0.2096              |

**Table 7** Non-COVID-19 and COVID-19 period

| Variables        | Non-COVID-19 period |                     |                     |                     | COVID-19 period     |                     |                     |                     |
|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                  | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                |
| Age              | -0.0711*** (0.0225) | -0.0684*** (0.0227) | -0.0189*** (0.0051) | -0.0184*** (0.0052) | -0.0080** (0.0039)  | -0.0080** (0.0040)  | -0.0080** (0.0039)  | -0.0080** (0.0040)  |
| Board_experience |                     |                     |                     |                     | 0.0046 (0.0080)     | 0.0036 (0.0081)     | 0.0046 (0.0080)     | 0.0036 (0.0081)     |
| Tenure           |                     |                     |                     |                     | -0.0308*** (0.0034) | -0.0368*** (0.0034) | -0.0308*** (0.0034) | -0.0368*** (0.0034) |
| Education        | 0.0023 (0.0080)     | 0.0015 (0.0082)     | 0.0034 (0.0079)     | 0.0025 (0.0081)     | -0.0361*** (0.0060) | -0.0370*** (0.0061) | -0.0361*** (0.0060) | -0.0370*** (0.0061) |
| Firm_size        | -0.0306*** (0.0033) | -0.0366*** (0.0034) | -0.0299*** (0.0033) | -0.0360*** (0.0033) | -0.0342 (0.0225)    | -0.0309 (0.0230)    | -0.0342 (0.0225)    | -0.0309 (0.0230)    |
| MTB              | -0.0338*** (0.0060) | -0.0347*** (0.0060) | -0.0337*** (0.0060) | -0.0346*** (0.0061) | 0.0098* (0.0054)    | 0.0086 (0.0053)     | 0.0098* (0.0054)    | 0.0086 (0.0053)     |
| Tang             | -0.0301 (0.0222)    | -0.0268 (0.0228)    | -0.0310 (0.0224)    | -0.0276 (0.0229)    | 0.0405 (0.0246)     | 0.0443* (0.0247)    | 0.0405 (0.0246)     | 0.0443* (0.0247)    |
| Sales            | 0.0087 (0.0054)     | 0.0075 (0.0054)     | 0.0083 (0.0053)     | 0.0071 (0.0052)     | -0.0315 (0.0371)    | 0.1216*** (0.0359)  | -0.0315 (0.0371)    | 0.1216*** (0.0359)  |
| Lev              | 0.0371 (0.0246)     | 0.0409* (0.0248)    | 0.0339 (0.0245)     | 0.0378 (0.0247)     | 1.2726*** (0.0953)  | 1.4520*** (0.0965)  | 1.2726*** (0.0953)  | 1.4520*** (0.0965)  |
| RMKT             | -0.0358 (0.0370)    | 0.1181*** (0.0358)  | -0.0429 (0.0369)    | 0.1110*** (0.0358)  | 3.358               | 3.358               | 3.358               | 3.358               |
| Constant         | 1.5329*** (0.1308)  | 1.7010*** (0.1317)  | 1.2863*** (0.0932)  | 1.4643*** (0.0944)  | 0.1752              | 0.2022              | 0.1752              | 0.2022              |
| Observations     | 3,368               | 3,368               | 3,369               | 3,369               |                     |                     |                     |                     |
| R-squared        | 0.1762              | 0.2028              | 0.1798              | 0.2062              |                     |                     |                     |                     |

**Note(s):** The table shows the regression results of [equation \(1\)](#) for two sub-periods: non-COVID-19 and COVID-19 time. CEO experience is measured by CEO age (Age), time on board of the CEO (Board\_experience) and CEO tenure (Tenure), while risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\*, and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

(continued)

Table 7

| Variables        | COVID-19 period     |                     |                     |                     | IDIO                |
|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                  | STD                 | IDIO                | STD                 | IDIO                | STD                 |
| Age              | -0.1147*** (0.0324) | -0.1125*** (0.0329) | -0.0293*** (0.0056) | -0.0303*** (0.0056) | -0.0218*** (0.0054) |
| Board_experience |                     |                     |                     |                     | -0.0185* (0.0111)   |
| Tenure           |                     |                     |                     |                     | -0.0276*** (0.0042) |
| Education        | -0.0191* (0.0109)   | -0.0185* (0.0110)   | -0.0171 (0.0111)    | -0.0167 (0.0111)    | -0.0166*** (0.0042) |
| Firm_size        | -0.0169*** (0.0042) | -0.0279*** (0.0043) | -0.0156*** (0.0042) | -0.0266*** (0.0042) | -0.0503*** (0.0098) |
| MTB              | -0.0497*** (0.0097) | -0.0502*** (0.0100) | -0.0480*** (0.0096) | -0.0483*** (0.0099) | -0.0316 (0.0292)    |
| Tang             | -0.0319 (0.0291)    | -0.0307 (0.0293)    | -0.0230 (0.0288)    | -0.0208 (0.0288)    | 0.0204* (0.0116)    |
| Sales            | 0.0183 (0.0112)     | 0.0169 (0.0113)     | 0.0183* (0.0109)    | 0.0169 (0.0109)     | 0.0336 (0.0330)     |
| Lev              | 0.0243 (0.0335)     | 0.0409 (0.0336)     | 0.0279 (0.0334)     | 0.0445 (0.0333)     | 0.4693*** (0.1038)  |
| RMKT             | 0.4762*** (0.1020)  | 0.5028*** (0.1018)  | 0.4760*** (0.1034)  | 0.5029*** (0.1030)  | 1.0183*** (0.1103)  |
| Constant         | 1.4447*** (0.1704)  | 1.7085*** (0.1721)  | 1.0264*** (0.1101)  | 1.3003*** (0.1112)  | 982                 |
| Observations     | 981                 | 981                 | 983                 | 983                 | 982                 |
| R-squared        | 0.1677              | 0.2091              | 0.1851              | 0.2280              | 0.1706              |
|                  |                     |                     |                     |                     | 0.2148              |

By using system-GMM, the [equation \(1\)](#) is re-estimated with the presence of lagged risk-taking. As presented in [Table 8](#), the CEO experience coefficients remain negative and statistically significant in all columns. These outcomes are consistent with previous findings, and support the *H1a*. The *p*-values of the Hansen test and AR2 test confirm the validity of instruments and the absence of serial autocorrelation, correspondingly.

As an extension, we implement a difference-in-difference (DID) framework leveraging CEO succession events to further validate the finding on relationship between CEO experience and firm risk-taking. Specifically, we define the treatment group as firms that experienced a CEO transition from a high-risk profile to a low-risk profile (risk profile is measured by ROA volatility, and high/low is separated based on median comparison). To ensure comparability, we use propensity score matching to construct a control group of firms with similar characteristics.

We then examine the changes in firm-level risk-taking before and after the CEO transition from high- to low-risk profile. The DID regression results, as shown in [Table 9](#), indicate that the coefficients of CEO age, experience and tenure remain negative and statistically significant, confirming our main findings that more experienced CEOs are associated with reduced firm risk-taking. Meanwhile, the coefficients of Turnover (denoted to CEO turnover) are positive and statistically significant in all columns, suggesting that changes in CEO role can increase firm risk. Specially, coefficients of the interaction term for the Treat\_group (=Turnover\*Hightolow\_Risk) are negative, implying that firms with high-to-low risk CEO transitions exhibit a reduction in risk. However, this effect is not very strong. This result strengthens the robustness of our findings by confirming that CEO experience independently contributes to risk reduction, beyond what can be attributed solely to CEO succession events involving changes in risk profiles.

In addition, we present the regression results for [equation \(1\)](#) when all three measures of experience are included in [Appendix 3](#). The results indicate that CEO age and board experience continue to have a detrimental effect on corporate risk-taking.

As an extension, to investigate the combined influence of CEO experience and the external environment, we incorporate interaction terms between CEO characteristics and external environmental variables into the model:

$$\begin{aligned} \text{Risk\_taking}_{it} = & \alpha_0 + \alpha_1 \text{CEO\_experience}_{it-1} + \alpha_2 \text{Education}_{it-1} + \alpha_3 \text{Ext\_environment}_{it-1} \\ & + \beta \text{CEO\_experience}_{it-1} * \text{Ext\_environment}_{it-1} \\ & + \gamma \text{Education}_{it-1} * \text{Ext\_environment}_{it-1} + \delta \text{Controls}_{it-1} + \varepsilon_{it} \end{aligned} \quad (4)$$

where Ext\_environment is Voice and Accountability, Political stability and Absence of Violence; Government Effectiveness; Regulatory quality; Control of Corruption; Rule of Law; which are drawn from Worldwide Governance Indicators (WGIs) database of World Bank; and Quality of Government which is the aggregate of these six indicators.

The results for the Quality of Government are provided in [Appendix 4](#) (results for the individual indicators are untabulated to save space). Our findings show that a stronger external environment tends to encourage firms to take more risk. Interestingly, the coefficients of the interaction term between CEO experience and Quality of Government are negative and statistically significant, indicating that the positive influence of the external environment on risk-taking is weaker in firms managed by CEOs with greater board experience and longer tenure.

### 4.3 CEO experience and firm risk-taking over business life cycle

Depending on a company's life cycle, the kind of markets it serves and the level of competition, even its goals may vary. Therefore, the significance of CEO experience in

**Table 8** System-GMM

| Variables                 | STD                 | IDIO                                      | STD                 | IDIO                                      | STD                 | IDIO                |
|---------------------------|---------------------|-------------------------------------------|---------------------|-------------------------------------------|---------------------|---------------------|
| STD                       | 0.5217*** (0.0285)  |                                           | 0.5129*** (0.0286)  |                                           | 0.5158*** (0.0286)  |                     |
| IDIO                      |                     | 0.5467*** (0.0283)<br>−0.0400*** (0.0134) |                     | 0.5396*** (0.0284)<br>−0.0115*** (0.0029) |                     | 0.5418*** (0.0284)  |
| Age                       | −0.0422*** (0.0136) |                                           | −0.0121*** (0.0029) |                                           |                     |                     |
| Board_experience          |                     |                                           |                     |                                           |                     |                     |
| Tenure                    |                     |                                           |                     |                                           | −0.0060*** (0.0022) | −0.0060*** (0.0021) |
| Education                 | −0.0018 (0.0041)    | −0.0015 (0.0040)                          | −0.0012 (0.0041)    | −0.001 (0.0040)                           | −0.0006 (0.0041)    | −0.0004 (0.0040)    |
| Firm_size                 | −0.0138*** (0.0021) | −0.0165*** (0.0022)                       | −0.0138*** (0.0021) | −0.0164*** (0.0022)                       | −0.0141*** (0.0022) | −0.0167*** (0.0023) |
| MTB                       | −0.0225*** (0.0036) | −0.0226*** (0.0036)                       | −0.0222*** (0.0037) | −0.0223*** (0.0036)                       | −0.0235*** (0.0038) | −0.0235*** (0.0037) |
| Tang                      | −0.0129 (0.0116)    | −0.0111 (0.0113)                          | −0.0127 (0.0117)    | −0.0108 (0.0114)                          | −0.0148 (0.0119)    | −0.0129 (0.0116)    |
| Sales                     | 0.004 (0.0041)      | 0.0027 (0.0041)                           | 0.0045 (0.0041)     | 0.0028 (0.0040)                           | 0.0056 (0.0042)     | 0.0038 (0.0041)     |
| Lev                       | 0.0355*** (0.0135)  | 0.0356*** (0.0132)                        | 0.0341*** (0.0135)  | 0.0343*** (0.0131)                        | 0.0387*** (0.0137)  | 0.0387*** (0.0134)  |
| Observations              | 4,349               | 4,349                                     | 4,352               | 4,352                                     | 4,340               | 4,340               |
| AR1 ( <i>p</i> -value)    | 0.0000              | 0.0000                                    | 0.0000              | 0.0000                                    | 0.0000              | 0.0000              |
| AR2 ( <i>p</i> -value)    | 0.2238              | 0.9559                                    | 0.1803              | 0.8592                                    | 0.1687              | 0.8398              |
| Hansen ( <i>p</i> -value) | 0.0531              | 0.0617                                    | 0.0545              | 0.0668                                    | 0.0532              | 0.0675              |

**Note(s):** The table shows the regression results of [equation \(1\)](#) used by the system-GMM method. CEO experience is measured by CEO age (Age), time on board of the CEO (Board\_experience) and CEO tenure (Tenure) while risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\*, and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

**Table 9** Difference-in-difference regression results

| Variables        | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                |
|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Hightlow_Risk    | −0.0048 (0.0088)    | −0.0051 (0.0090)    | −0.0038 (0.0088)    | −0.004 (0.0090)     | −0.0043 (0.0089)    | −0.0046 (0.0091)    |
| Turnover         | 0.0256*** (0.0073)  | 0.0257*** (0.0073)  | 0.0209*** (0.0072)  | 0.0209*** (0.0073)  | 0.0225*** (0.0074)  | 0.0222*** (0.0075)  |
| Treat_group      | −0.0167 (0.0105)    | −0.017 (0.0106)     | −0.0172* (0.0104)   | −0.0176* (0.0106)   | −0.0153 (0.0106)    | −0.0156 (0.0108)    |
| Age              | −0.0749** (0.0212)  | −0.0726*** (0.0216) |                     |                     |                     |                     |
| Board_experience |                     |                     | −0.0203*** (0.0045) | −0.0204*** (0.0045) |                     |                     |
| Tenure           |                     |                     |                     |                     | −0.0083** (0.0036)  | −0.0088** (0.0036)  |
| Education        | −0.0042 (0.0074)    | −0.0046 (0.0076)    | −0.0027 (0.0074)    | −0.0032 (0.0075)    | −0.0019 (0.0074)    | −0.0024 (0.0075)    |
| Firm_size        | −0.0272*** (0.0031) | −0.0345*** (0.0032) | −0.0264*** (0.0031) | −0.0337*** (0.0031) | −0.0272*** (0.0032) | −0.0346*** (0.0032) |
| MTB              | −0.0383*** (0.0056) | −0.0390*** (0.0057) | −0.0381*** (0.0057) | −0.0387*** (0.0058) | −0.0402*** (0.0058) | −0.0408*** (0.0059) |
| Tang             | −0.0258 (0.0211)    | −0.023 (0.0216)     | −0.0256 (0.0212)    | −0.0227 (0.0218)    | −0.0303 (0.0214)    | −0.0276 (0.0219)    |
| Sales            | 0.0105** (0.0050)   | 0.0095* (0.0050)    | 0.0101** (0.0050)   | 0.0090* (0.0049)    | 0.0117** (0.0051)   | 0.0106** (0.0050)   |
| Lev              | 0.0315 (0.0230)     | 0.0387* (0.0231)    | 0.0295 (0.0228)     | 0.0367 (0.0230)     | 0.0356 (0.0231)     | 0.0429* (0.0232)    |
| RMKT             | 0.1906*** (0.0169)  | 0.2315*** (0.0169)  | 0.1804*** (0.0167)  | 0.2212*** (0.0167)  | 0.1898*** (0.0170)  | 0.2301*** (0.0170)  |
| Constant         | 1.5410*** (0.1208)  | 1.7067*** (0.1228)  | 1.2778*** (0.0839)  | 1.4528*** (0.0854)  | 1.2616*** (0.0860)  | 1.4379*** (0.0876)  |
| Observations     | 4,349               | 4,349               | 4,352               | 4,352               | 4,340               | 4,340               |
| R-squared        | 0.182               | 0.209               | 0.186               | 0.215               | 0.179               | 0.208               |

**Note(s):** The table shows the difference-in-difference regression results. Treatment group includes firms that experienced a CEO transition from a high-risk profile to a low-risk profile (Risk profile is measured by ROA volatility, and high/low is separated based on median comparison). The method of propensity score matching is used to construct a control group of firms with similar characteristics. CEO experience is measured by CEO age, board experience and tenure, respectively. Risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\* and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

relation to risk-taking may change based on the business stages. We follow recent studies, e.g. [Castro et al. \(2016\)](#), [Habib and Hasan \(2017\)](#) and [Shahzad et al. \(2019\)](#), to depend on the cash-flow approach proposed by [Dickinson \(2011\)](#) (see [Appendix 5](#)) to determine which stage a firm is in.

In practice, some common proxies used in corporate life cycle literature to identify company stages are firm age, size and sales growth. However, these indicators ignore the possibility of cyclicalities in the corporate life cycle, and their linear pattern has led to criticism of their use ([Dickinson, 2011](#)). The signs of the operating, investing and financing cash flows revealed in the cash flow statement, in contrast to these proxies, can indicate the changes in firm strategy, resource allocation and operating capability over different life stages. Particularly in the context of risk-taking, the cash-flow pattern is more reliable indicator of business stages because it provides insight into the way firms respond to both internal and external business environment's changes. In addition, the cash flow classification is shown to be strongly consistent with the economic theory of the life cycle (see [Dickinson, 2011](#)).

As shown in [Appendix 6](#), the means of STD and IDIO are higher for firms in the introduction and decline stages, suggesting greater risk-taking. Leverage is highest during the introduction and growth stages, supporting the notion that firms in early stages rely heavily on external finance. Profitability is highest during the maturity stage and lowest in decline, while asset tangibility increases from introduction to maturity, then drops sharply in the decline stage, indicating that matured firms hold more fixed assets, while declining firms may have to liquid assets. These patterns support the validity of Dickinson's classification in the context of Vietnamese firms.

To test the changes in the relationship between CEO experience and corporate risk-taking during different life cycle stages, we estimate the [equation \(3\)](#) and present the results in [Table 10](#). As shown in the table, the coefficients of CEO age, board experience and tenure remain negative and statistically significant in all columns. Interestingly, the coefficients of interaction terms: *Decline\_dummy\*CEO\_experience* and *Decline\_dummy\*Education* are significant negatively, suggesting that highly educated and experienced CEOs can assist declining firms in lowering risk to a notable extent.

To further ensure about the robustness of findings, we use firm age as an alternative method to identify business stages. While 5 is commonly used to characterize young enterprises (see [Coad et al., 2018](#); [Fort et al., 2013](#)), [Brown and Medoff \(2003\)](#) and [Coad et al. \(2016\)](#) suggest thresholds of 15 and 30 to define other stages. Besides, sales growth is frequently used as an indicator for classifying firms into different stages of the business life cycle ([Anthony and Ramesh, 1992](#); [Dickinson, 2011](#)). According to [Yan and Zhao \(2010\)](#), a sales growth rate of 15% serves as a threshold to distinguish between growth firms (sales growth  $\geq 15\%$ ) and mature firms (sales growth  $< 15\%$ ). Negative sales growth, in contrast, is commonly associated with firms in the decline stage. As shown in [Appendices 7](#) and [8](#), these two alternative classification methods give consistent results with those presented in [Table 10](#).

## 5. Conclusion

Using a comprehensive sample of listed firms on Vietnamese stock exchanges during the period of 2010–2021, our study provides evidence that experienced CEOs are associated with lower level of firm risk-taking. The result supports the stewardship theory and suggests that CEOs with high experience (i.e. life experience, board experience, position experience), as stewards of the firm, tend to act in the long-term interests of the organization rather than being self-interested. The finding reinforces the idea that a CEO with substantial experience is better equipped to make decisions that prioritize the company's overall well-being, including managing risk in a responsible manner. The risk-taking lowering effect of experienced CEOs can be found across all stages of the firm life cycle, emphasizing the

**Table 10** CEO experience and risk-taking over 4 stages of business life

| Variables                    | Age                 |                     | Board_experience    |                     | Tenure              |                     |
|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                              | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                |
| CEO_experience               | -0.0888*** (0.0209) | -0.0866*** (0.0208) | -0.0150*** (0.0032) | -0.0154*** (0.0032) | -0.0087** (0.0038)  | -0.0091** (0.0038)  |
| CEO_experience*Growth_dummy  | 0.0073 (0.0372)     | 0.0057 (0.0370)     | -0.0106 (0.0070)    | -0.0120* (0.0069)   | -0.0013 (0.0068)    | -0.0026 (0.0067)    |
| CEO_experience*Mature_dummy  | 0.0144 (0.0304)     | 0.0167 (0.0302)     | -0.0147*** (0.0053) | -0.0140*** (0.0053) | -0.0050 (0.0054)    | -0.0037 (0.0054)    |
| CEO_experience*Decline_dummy | -0.0065 (0.0411)    | -0.0081 (0.0409)    | -0.0159** (0.0079)  | -0.0159** (0.0079)  | -0.0157** (0.0076)  | -0.0151** (0.0076)  |
| Education                    | 0.0057 (0.0074)     | 0.0062 (0.0074)     | 0.0097 (0.0079)     | 0.0100 (0.0078)     | 0.0096 (0.0073)     | 0.0100 (0.0073)     |
| Education*Growth_dummy       | -0.0186 (0.0131)    | -0.0228* (0.0130)   | -0.0242* (0.0136)   | -0.0275** (0.0135)  | -0.0215 (0.0131)    | -0.0260** (0.0130)  |
| Education*Mature_dummy       | -0.0039 (0.0103)    | -0.0036 (0.0102)    | -0.0048 (0.0108)    | -0.0045 (0.0108)    | -0.0076 (0.0102)    | -0.0071 (0.0102)    |
| Education*Decline_dummy      | -0.0307** (0.0148)  | -0.0329** (0.0147)  | -0.0380** (0.0158)  | -0.0392** (0.0157)  | -0.0310** (0.0147)  | -0.0330** (0.0146)  |
| Growth_dummy                 | -0.0013 (0.1509)    | 0.0138 (0.1502)     | 0.0688* (0.0361)    | 0.0794** (0.0360)   | 0.0355 (0.0328)     | 0.0478 (0.0327)     |
| Mature_dummy                 | -0.0584 (0.1237)    | -0.0673 (0.1231)    | 0.0379 (0.0292)     | 0.0357 (0.0291)     | 0.0134 (0.0257)     | 0.0111 (0.0256)     |
| Decline_dummy                | 0.1052 (0.1672)     | 0.1173 (0.1664)     | 0.1288*** (0.0417)  | 0.1329*** (0.0416)  | 0.1074*** (0.0364)  | 0.1119*** (0.0362)  |
| Firm_size                    | -0.0239*** (0.0015) | -0.0314*** (0.0015) | -0.0245*** (0.0016) | -0.0315*** (0.0016) | -0.0240*** (0.0015) | -0.0315*** (0.0015) |
| MTB                          | -0.0407*** (0.0032) | -0.0397*** (0.0032) | -0.0399*** (0.0035) | -0.0388*** (0.0034) | -0.0424*** (0.0033) | -0.0413*** (0.0032) |
| Tang                         | -0.0317*** (0.0113) | -0.0296*** (0.0112) | -0.0308*** (0.0119) | -0.0286*** (0.0119) | -0.0360*** (0.0113) | -0.0340*** (0.0113) |
| Sales                        | 0.0079* (0.0044)    | 0.0071 (0.0044)     | 0.0072 (0.0047)     | 0.0061 (0.0047)     | 0.0090** (0.0044)   | 0.0081* (0.0044)    |
| Lev                          | 0.0361*** (0.0126)  | 0.0449*** (0.0125)  | 0.0284** (0.0132)   | 0.0363*** (0.0131)  | 0.0413*** (0.0126)  | 0.0501*** (0.0125)  |
| RMKT                         | 0.0645*** (0.0120)  | 0.0823*** (0.0119)  | 0.0540*** (0.0119)  | 0.0727*** (0.0119)  | 0.0649*** (0.0120)  | 0.0825*** (0.0119)  |
| Constant                     | 1.4587*** (0.0937)  | 1.6331*** (0.0933)  | 1.1550*** (0.0451)  | 1.3267*** (0.0450)  | 1.1214*** (0.0428)  | 1.3061*** (0.0426)  |
| Observations                 | 4,349               | 4,349               | 3,863               | 3,863               | 4,340               | 4,340               |
| R-squared                    | 0.1392              | 0.1781              | 0.1493              | 0.1872              | 0.1389              | 0.1781              |

**Note(s):** The table shows the regression results of [equation \(1\)](#) with the existence of stage\_dummy variables. CEO experience is measured by CEO age, board experience and tenure, respectively. Risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\*, and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

importance of CEO experience in managing and mitigating risks. Further, during the stage of decline, the importance of highly educated and experienced CEOs are more pronounced. The evidence implies that firms, particularly those in the most challenging phases of their life, may benefit from selecting CEOs with strong educational and working backgrounds that provide them with frameworks for making informed and less risky decisions. In addition, the results have some implications for investors and regulators when defining CEO experience as a useful signal of managerial quality and risk discipline. Investors should incorporate CEO background information when evaluating firm risk profiles, while policymakers in emerging markets can view experienced CEOs as an effective governance mechanism, especially where external monitoring is weaker.

Some limitations of this study should be acknowledged. First, the sample is restricted to listed Vietnamese companies, so caution is needed when generalizing the findings to other emerging markets. Second, further research is warranted, as the manual-collected data set does not capture all dimensions of CEO experience, such as specific areas of specialization or skill sets.

## Notes

- [1]. The means of Age, Board\_experience and Tenure presented in [Table 1](#) are the means of their natural logarithm values.
- [2]. The same issue can be found in [Orens and Reheul \(2013\)](#), [Wang et al. \(2016\)](#). Besides treating the three CEO experience measures separately, we also conduct tests with all three measures and present results in [Appendix 3](#).
- [3]. Most of SOEs in our sample are in Industrials (44.64%), Basic materials (19.63%) and Consumer Goods (11.36%).
- [4]. Approximately 71% of non-SOEs in the sample have foreign ownership, with an average level of 14% (ranging from 1% to 49%).

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## Appendix 1

**Table A1** Variable definitions

| <i>Variables</i>             | <i>Definition</i>    | <i>Measurement</i>                                                                                                                  |
|------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| STD                          | Risk-taking          | Annualized standard deviation of daily stock return                                                                                 |
| IDIO                         | Risk-taking          | Annualized standard deviation of the residuals derived from the market model [equation (2)]                                         |
| Age                          | CEO age              | Ln(CEO age)                                                                                                                         |
| Board_experience             | CEO board experience | Ln(1 + number of years the CEO has served on board)                                                                                 |
| Tenure                       | CEO tenure           | Ln(1 + number of years the CEO has served in the position)                                                                          |
| Education                    | CEO education        | 1 if CEO qualifications lower than bachelor; 2 if they have bachelor degree as highest qualification; and 3 if higher than bachelor |
| Duality                      | Duality              | 1 if CEOs are chairmen and 0 otherwise                                                                                              |
| Board_size                   | Board size           | Number of board members                                                                                                             |
| Independent                  | Independent ratio    | Number of independent members in boards                                                                                             |
| Firm_size                    | Firm size            | Ln(total assets)                                                                                                                    |
| MTB                          | Market-to-book ratio | Market value of equity/book value of equity                                                                                         |
| Tang                         | Tangibility          | Fixed assets/total assets                                                                                                           |
| Sales                        | Sales growth         | (Sales-lagged sales)/lagged sales                                                                                                   |
| Lev                          | Leverage             | Total liabilities/total assets                                                                                                      |
| RMKT                         | Market return        | Annual return on VN-Index                                                                                                           |
| Source(s): Authors' own work |                      |                                                                                                                                     |

## Appendix 2

**Table A2** Test of difference between the coefficients of CEO experience

| <i>Variables</i>                                                                                                                                                                                               | <i>STD</i> | <i>IDIO</i> | <i>STD</i> | <i>IDIO</i> | <i>STD</i> | <i>IDIO</i> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|------------|-------------|------------|-------------|
| <i>Panel A</i>                                                                                                                                                                                                 |            |             |            |             |            |             |
| <i>Female vs male CEOs</i>                                                                                                                                                                                     |            |             |            |             |            |             |
| Age                                                                                                                                                                                                            | 2.03**     | 2.12**      |            |             |            |             |
| Board_experience                                                                                                                                                                                               |            |             | 2.06**     | 2.12**      |            |             |
| Tenure                                                                                                                                                                                                         |            |             |            |             | 1.63       | 1.65*       |
| <i>Panel B</i>                                                                                                                                                                                                 |            |             |            |             |            |             |
| <i>SOEs vs non-SOEs</i>                                                                                                                                                                                        |            |             |            |             |            |             |
| Age                                                                                                                                                                                                            | 2.68***    | 2.43***     |            |             |            |             |
| Board_experience                                                                                                                                                                                               |            |             | 2.67***    | 2.45***     |            |             |
| Tenure                                                                                                                                                                                                         |            |             |            |             | 2.70***    | 2.37***     |
| <i>Panel C</i>                                                                                                                                                                                                 |            |             |            |             |            |             |
| <i>Non-COVID-19 vs COVID-19 period</i>                                                                                                                                                                         |            |             |            |             |            |             |
| Age                                                                                                                                                                                                            | 13.5***    | 11.64***    |            |             |            |             |
| Board_experience                                                                                                                                                                                               |            |             | 12.23***   | 10.47***    |            |             |
| Tenure                                                                                                                                                                                                         |            |             |            |             | 13.22***   | 11.37***    |
| Note(s): This table provides the results of Chow test of difference between the coefficients of CEO experience across subsamples. *, ** and *** indicate significance at the 10, 5 and 1% levels, respectively |            |             |            |             |            |             |
| Source(s): Authors' own work                                                                                                                                                                                   |            |             |            |             |            |             |

**Table A3** Regression results with all measures of CEO experience

| Variables        | All firms           |                     | Non-crisis          |                     | COVID-19            |                     |
|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                  | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                |
| Age              | -0.0417*** (0.0146) | -0.0379*** (0.0146) | -0.0479*** (0.0164) | -0.0441*** (0.0163) | -0.0590* (0.0310)   | -0.0523* (0.0311)   |
| Board_experience | -0.0222*** (0.0032) | -0.0223*** (0.0032) | -0.0172*** (0.0038) | -0.0172*** (0.0038) | -0.0205*** (0.0063) | -0.0220*** (0.0063) |
| Tenure           | -0.0006 (0.0029)    | -0.0006 (0.0029)    | 0.0007 (0.0032)     | 0.0009 (0.0032)     | -0.0089 (0.0063)    | -0.0093 (0.0063)    |
| Education        | -0.0035 (0.0044)    | -0.004 (0.0044)     | 0.0025 (0.0049)     | 0.0018 (0.0049)     | -0.0220** (0.0093)  | -0.0223** (0.0093)  |
| Constant         | 1.3400*** (0.0677)  | 1.5129*** (0.0673)  | 1.4663*** (0.0758)  | 1.5953*** (0.0756)  | 1.2179*** (0.1444)  | 1.4879*** (0.1448)  |
| Observations     | 4,337               | 4,337               | 3,357               | 3,357               | 980                 | 980                 |
| R-squared        | 0.152               | 0.191               | 0.166               | 0.198               | 0.189               | 0.233               |

**Note(s):** The table shows the regression results of [equation \(1\)](#) with the existence of stage\_dummy variables. CEO experience is measured by CEO age, time on board and tenure while risk-taking is measured by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Control variables include board size (Board\_size), Duality (Duality), Independent (Independent), firm size (Firm\_size), Market to book ratio (MTB), Tangibility (Tang), Sales growth (Sales), Leverage (Lev) and Market return (RMKT), but are not shown to save space. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\*, and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

**Table A4** The role of external environment

| Variables                              | STD                | IDIO               | STD                 | IDIO                | STD                 | IDIO                |
|----------------------------------------|--------------------|--------------------|---------------------|---------------------|---------------------|---------------------|
| Age                                    | -0.1514** (0.0708) | -0.1819* (0.0726)  |                     |                     |                     |                     |
| Age*quality of government              | -0.0201 (0.0271)   | -0.0331 (0.0278)   |                     |                     |                     |                     |
| Board_experience                       |                    |                    | -0.0554*** (0.0145) | -0.0622*** (0.0149) |                     |                     |
| Board_experience*quality of government |                    |                    | -0.0111* (0.0057)   | -0.0135** (0.0059)  |                     |                     |
| Tenure                                 |                    |                    |                     |                     | -0.0410*** (0.0132) | -0.0476*** (0.0135) |
| Tenure*quality of government           |                    |                    |                     |                     | -0.0114** (0.0051)  | -0.0139*** (0.0052) |
| Quality of government                  | 0.1219 (0.1101)    | 0.1666 (0.1128)    | 0.0642* (0.0272)    | 0.0642* (0.0279)    | 0.0613** (0.0242)   | 0.0596** (0.0248)   |
| Education                              | -0.0413* (0.0249)  | -0.0323 (0.0255)   | -0.0344 (0.0246)    | -0.0248 (0.0252)    | -0.0386 (0.0249)    | -0.0293 (0.0255)    |
| Education*quality of government        | -0.0115 (0.0095)   | -0.0067 (0.0097)   | -0.0095 (0.0094)    | -0.0045 (0.0096)    | -0.0115 (0.0095)    | -0.0066 (0.0097)    |
| GDP                                    | -0.0047** (0.0022) | -0.0046** (0.0022) | -0.0047** (0.0021)  | -0.0047** (0.0022)  | -0.0046** (0.0022)  | -0.0046** (0.0022)  |
| Constant                               | 1.2176*** (0.2878) | 1.3033*** (0.2949) | 0.7466*** (0.0700)  | 0.7293*** (0.0716)  | 0.6879*** (0.0631)  | 0.6648*** (0.0646)  |
| Observations                           | 4,754              | 4,754              | 4,757               | 4,757               | 4,744               | 4,744               |
| R-squared                              | 0.071              | 0.063              | 0.083               | 0.077               | 0.066               | 0.059               |

**Note(s):** The table shows the regression results of [equation \(4\)](#). CEO experience is measured by CEO age, time on board and tenure while risk-taking is measured by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Quality of Government is the total of Voice and Accountability, Political stability and Absence of Violence, Government Effectiveness, Regulatory quality, Control of corruption, and Rule of Law, which are drawn from Worldwide Governance Indicators (WGIs) database of World Bank. Control variables include board size (Board\_size), Duality (Duality), Independent (Independent), firm size (Firm\_size), Market to book ratio (MTB), Tangibility (Tang), Sales growth (Sales), Leverage (Lev), Market return (RMKT) and GDP growth rate; but all, except GDP, are not shown to save space. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\*, and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

## Appendix 5

**Table A5** Cash-flow signs as a proxy for business life cycle

| <i>Cash flow</i> | <i>Introduction</i> | <i>Growth</i> | <i>Maturity</i> | <i>Decline</i> |
|------------------|---------------------|---------------|-----------------|----------------|
| Operating        | –                   | +             | +               | –              |
| Investing        | –                   | –             | –               | +              |
| Financing        | +                   | +             | –               | –/+            |

Source(s): Dickinson (2011, p. 1972)

## Appendix 6

**Table A6** Distribution of firms across life cycle stages

| <i>Stages</i>       | <i>Introduction</i> | <i>Growth</i> | <i>Maturity</i> | <i>Decline</i> |
|---------------------|---------------------|---------------|-----------------|----------------|
| No. of observations | 715                 | 812           | 1,709           | 497            |
| Percent             | 19.15%              | 21.75%        | 45.78%          | 13.31%         |
| NCF from operating  | –1.35E + 11         | 2.93E + 11    | 3.48E + 11      | –1.06E + 11    |
| NCF from investing  | –1.73E + 11         | –6.32E + 11   | –1.47E + 11     | 1.03E + 11     |
| NCF from financing  | 2.90E + 11          | 4.19E + 11    | –2.05E + 11     | –1.06E + 10    |
| STD                 | 0.4752              | 0.4416        | 0.4365          | 0.4896         |
| IDIO                | 0.4543              | 0.4195        | 0.4212          | 0.4738         |
| Profitability       | 0.0743              | 0.1086        | 0.1500          | 0.0590         |
| Tangibility         | 0.1594              | 0.2484        | 0.2805          | 0.1068         |
| Leverage            | 0.3401              | 0.3069        | 0.1861          | 0.1847         |

Source(s): Authors' own work

**Table A7** Firm age as an alternative measure of life cycle stages

| Variables                    | Age                 |                     | Board_experience    |                     | Tenure              |                     |
|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                              | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                |
| CEO_experience               | -0.0688 (0.0645)    | -0.0539 (0.0646)    | -0.0117*** (0.0037) | -0.0117*** (0.0037) | -0.0088 (0.0150)    | -0.0108 (0.0150)    |
| CEO_experience*Growth_dummy  | 0.0183 (0.0696)     | 0.0012 (0.0697)     | 0.0052 (0.0059)     | 0.0047 (0.0059)     | 0.0031 (0.0156)     | 0.0037 (0.0157)     |
| CEO_experience*Mature_dummy  | -0.0416 (0.0675)    | -0.0495 (0.0676)    | -0.0227*** (0.0052) | -0.0216*** (0.0052) | -0.0035 (0.0155)    | -0.0005 (0.0155)    |
| CEO_experience*Decline_dummy | -0.0059 (0.0681)    | -0.0229 (0.0682)    | -0.0176*** (0.0056) | -0.0192*** (0.0057) | -0.006 (0.0155)     | -0.0046 (0.0155)    |
| Education                    | 0.0168 (0.0244)     | 0.0146 (0.0245)     | 0.0182 (0.0240)     | 0.0149 (0.0240)     | 0.0183 (0.0243)     | 0.0145 (0.0244)     |
| Education*Growth_dummy       | -0.0162 (0.0254)    | -0.0134 (0.0254)    | -0.0189 (0.0250)    | -0.0152 (0.0250)    | -0.0161 (0.0253)    | -0.0119 (0.0253)    |
| Education*Mature_dummy       | -0.0084 (0.0256)    | -0.0093 (0.0256)    | -0.0043 (0.0253)    | -0.004 (0.0253)     | -0.0092 (0.0255)    | -0.0086 (0.0255)    |
| Education*Decline_dummy      | -0.0453* (0.0261)   | -0.0415 (0.0261)    | -0.0442* (0.0258)   | -0.0387 (0.0258)    | -0.0435* (0.0259)   | -0.0381 (0.0259)    |
| Growth_dummy                 | -0.0271*** (0.0016) | -0.0345*** (0.0016) | -0.0263*** (0.0017) | -0.0333*** (0.0017) | -0.0272*** (0.0016) | -0.0345*** (0.0016) |
| Mature_dummy                 | -0.0396*** (0.0033) | -0.0403*** (0.0033) | -0.0373*** (0.0036) | -0.0384*** (0.0036) | -0.0408*** (0.0033) | -0.0415*** (0.0033) |
| Decline_dummy                | -0.0313*** (0.0113) | -0.0293*** (0.0113) | -0.0267*** (0.0120) | -0.0242*** (0.0120) | -0.0349*** (0.0114) | -0.0329*** (0.0114) |
| Firm_size                    | 0.0104** (0.0044)   | 0.0095** (0.0044)   | 0.0064 (0.0047)     | 0.0055 (0.0047)     | 0.0115*** (0.0044)  | 0.0106** (0.0044)   |
| MTB                          | 0.0358*** (0.0125)  | 0.0428*** (0.0126)  | 0.0268** (0.0132)   | 0.0333* (0.0132)    | 0.0404*** (0.0126)  | 0.0472*** (0.0126)  |
| Tang                         | 0.1838*** (0.0209)  | 0.2238*** (0.0209)  | 0.1189*** (0.0240)  | 0.1613*** (0.0241)  | 0.1852*** (0.0209)  | 0.2246*** (0.0209)  |
| Sales                        | 0.1758 (0.2756)     | 0.1995 (0.2759)     | 0.0798 (0.0580)     | 0.0683 (0.0581)     | 0.0196 (0.0648)     | 0.0054 (0.0648)     |
| Lev                          | 0.0208 (0.2784)     | 0.0883 (0.2788)     | 0.0366 (0.0591)     | 0.0398 (0.0591)     | 0.0068 (0.0655)     | 0.0036 (0.0656)     |
| RMKT                         | 0.01 (0.2860)       | 0.069 (0.2864)      | 0.0614 (0.0607)     | 0.0521 (0.0608)     | 0.0675 (0.0664)     | 0.0565 (0.0664)     |
| Constant                     | 1.4975*** (0.2666)  | 1.6191*** (0.2669)  | 1.2125*** (0.0720)  | 1.3871*** (0.0722)  | 1.2464*** (0.0762)  | 1.4316*** (0.0763)  |
| Observations                 | 4,349               | 4,349               | 3,863               | 3,863               | 4,340               | 4,340               |
| R-squared                    | 0.180               | 0.207               | 0.179               | 0.207               | 0.179               | 0.206               |

**Note(s):** The table shows the regression results of [equation \(1\)](#) when firm age is used to define the lifecycle stages (Introduction: firm age <= 5; Growth: 5< firm age <=15; Maturity: 15< firm age <=30; Decline: 30< firm age). CEO experience is measured by CEO age, board experience and tenure, respectively. Risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\*, and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work

**Table A8** Sales growth as an alternative measure of life cycle stages

| Variables                    | Age                 |                     | Board_experience    |                     | Tenure              |                     |
|------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                              | STD                 | IDIO                | STD                 | IDIO                | STD                 | IDIO                |
| CEO_experience               | -0.0781*** (0.0123) | -0.0747*** (0.0123) | -0.0157*** (0.0029) | -0.0156*** (0.0029) | -0.0109*** (0.0034) | -0.0112*** (0.0034) |
| CEO_experience*Mature_dummy  | -0.0073 (0.0057)    | -0.0068 (0.0058)    | -0.0042 (0.0046)    | -0.0045 (0.0046)    | 0.0009 (0.0050)     | 0.001 (0.0050)      |
| CEO_experience*Decline_dummy | -0.0103* (0.0053)   | -0.0098* (0.0053)   | -0.0141*** (0.0041) | -0.0141*** (0.0041) | -0.0017 (0.0045)    | -0.0015 (0.0045)    |
| Education                    | -0.0133** (0.0064)  | -0.0138** (0.0064)  | -0.0075 (0.0051)    | -0.0087* (0.0051)   | -0.0021 (0.0046)    | -0.0031 (0.0046)    |
| Education*Mature_dummy       | 0.0086 (0.0097)     | 0.0081 (0.0098)     | 0.0012 (0.0056)     | 0.0018 (0.0056)     | -0.0043 (0.0042)    | -0.0042 (0.0042)    |
| Education*Decline_dummy      | -0.0083* (0.0050)   | -0.0083* (0.0050)   | -0.0051 (0.0053)    | -0.0056 (0.0053)    | -0.0095* (0.0050)   | -0.0094* (0.0050)   |
| Mature_dummy                 | 0.0043 (0.0047)     | 0.0048 (0.0047)     | 0.0076 (0.0049)     | 0.0073 (0.0050)     | 0.0040 (0.0047)     | 0.0045 (0.0047)     |
| Decline_dummy                | 0.0208* (0.0090)    | 0.0205* (0.0090)    | 0.0198*** (0.0050)  | 0.0202*** (0.0050)  | 0.0048 (0.0038)     | 0.0052 (0.0038)     |
| Firm_size                    | -0.0269*** (0.0015) | -0.0342*** (0.0015) | -0.0261*** (0.0016) | -0.0332*** (0.0016) | -0.0269*** (0.0015) | -0.0343*** (0.0015) |
| MTB                          | -0.0372*** (0.0032) | -0.0378*** (0.0032) | -0.0348*** (0.0034) | -0.0358*** (0.0034) | -0.0388*** (0.0032) | -0.0393*** (0.0032) |
| Tang                         | -0.0318*** (0.0107) | -0.0284*** (0.0107) | -0.0297*** (0.0112) | -0.0258*** (0.0113) | -0.0368*** (0.0107) | -0.0333*** (0.0107) |
| Lev                          | 0.0432*** (0.0118)  | 0.0499*** (0.0119)  | 0.0348*** (0.0124)  | 0.0409*** (0.0125)  | 0.0474*** (0.0119)  | 0.0540*** (0.0119)  |
| RMKT                         | 0.4170*** (0.0638)  | 0.4969*** (0.0639)  | 0.3179*** (0.1071)  | 0.4378*** (0.1073)  | 0.4076*** (0.0636)  | 0.4873*** (0.0637)  |
| Constant                     | 1.5332*** (0.0616)  | 1.6885*** (0.0616)  | 1.2307*** (0.0432)  | 1.3942*** (0.0432)  | 1.2247*** (0.0408)  | 1.3953*** (0.0408)  |
| Observations                 | 4,678               | 4,678               | 4,187               | 4,187               | 4,668               | 4,668               |
| R-squared                    | 0.180               | 0.210               | 0.176               | 0.207               | 0.178               | 0.209               |

**Note(s):** The table shows the regression results of [equation \(1\)](#) when sales growth is used to define the lifecycle stages (Growth: sales growth  $\geq 15\%$ ; Maturity:  $0\% \leq$  sales growth  $< 15\%$ ; Decline: sales growth  $< 0\%$ ). CEO experience is measured by CEO age, board experience and tenure, respectively. Risk-taking is denoted by stock volatility (STD) and idiosyncratic risk (IDIO), respectively. Independent variables enter the regression with a one-year lag. Robust standard errors are in parentheses. \*, \*\* and \*\*\* indicate significance at the 10, 5 and 1% levels, respectively. Variable definitions, measurement and data sources can be found in [Appendix 1](#)

**Source(s):** Authors' own work